

One important step remains. The Openfiler set-up has only the GRUB register *hda* as a boot drive; *hdb* should also be made bootable. To do this, use the following commands (apply GRUB commands corresponding to the detected hard disks; here, *hda* was detected as *sda* and *hdb* as *sdb*):

```
#grub
grub>device (hd0) /dev/sda
grub>root (hd0,0)
grub>setup (hd0)
grub>device (hd1) /dev/sdb
grub>root (hd1,0)
grub>setup (hd1)
grub>quit
#
```

Verifying the installation

You can remove one hard disk at a time, and check whether the system boots properly. Once up and running using a single disk, check under **Volumes—Software RAID**; you will see the ‘clean’ and ‘degraded’ volumes (which remain unsynchronised), as shown in Figure 3. After re-installing the removed disk, proceed to add the member to the RAID 1 array, for synchronisation to start. Then select **Add Member** from the ADD column; carefully check the partition sizes, and add the required RAID 1 member to the array. You can see the synchronisation process status (Figure 4) immediately after this. If a new hard disk is added, you will require to create the partitions and then add them to the RAID array.

The remaining set-up requires you to configure network access via **System—Network Setup**; create the Network Access Configuration to allow access to the full subnet.

Important tips

- Use only manual partitioning. Auto partitioning will use the full disk for the OS, leaving no space for data storage.
- Make all software RAID partitions primary partitions—check the ‘*Force primary*’ check-box.
- By default, logs are stored on the root (/) partition. If it runs out of space, the system stops functioning. To protect against this, a separate */var/log* partition is recommended.
- Do not configure swap partitions as either RAID type, or primary partitions.
- You must configure GRUB to allow booting from both disks—else, if *hda* fails, the system will not boot—and OS redundancy by RAID 1 will be useless.
- Further configuration steps, with screen-shots, are given in the August and September issues of *LINUX For You*. Please refer to those for more information.

Enabling FTP

In the Web GUI, create a user group: **Accounts—Administration—Group Administration**; create *ftpgroup*, overriding GID to 1001.

Figure 1: RAID partitions

Array	Level	Array Size	Device Size	State	Synchronisation	Message	Add	Used to	Exclude
/dev/md0	RAID-1	7.98 GB	7.98 GB	Clean	Synchronised	RAID-1 partitions are used	+	(the new /) mirrored	Exclude
/dev/md1	RAID-1	768.62 MB	768.62 MB	Active	Synchronised	RAID-1 partitions are used	+	(the new /) mirrored	Exclude
/dev/md2	RAID-1	259.75 MB	259.75 MB	Clean	Synchronised	RAID-1 partitions are used	+	(the new /) mirrored	Exclude

Create a new RAID array

An existing RAID partition was found, or all existing RAID partitions are used. If you can [create new RAID partitions](#).

Figure 2: Verifying RAID 1 partitions

Array	Level	Array Size	Device Size	State	Synchronisation	Message	Add	Used to	Exclude
/dev/md0	RAID-1	7.98 GB	7.98 GB	Clean & Sync	Not started	RAID-1 partitions are used	+	(the new /) mirrored	Exclude
/dev/md1	RAID-1	768.62 MB	768.62 MB	Clean & Sync	Not started	RAID-1 partitions are used	+	(the new /) mirrored	Exclude
/dev/md2	RAID-1	259.75 MB	259.75 MB	Clean & Sync	Not started	RAID-1 partitions are used	+	(the new /) mirrored	Exclude
/dev/md3	RAID-1	7.98 GB	7.98 GB	Clean & Sync	Not started	RAID-1 partitions are used	+	(the new /) mirrored	Exclude

Create a new RAID array

An existing RAID partition was found, or all existing RAID partitions are used. If you can [create new RAID partitions](#).

Figure 3: RAID partitions with *sda* removed

Then create the *ftpuer1* user in **Accounts—Administration—User Administration**, overriding UID to 501. Go to **Services—FTP Server** and click **Enable**.

After you add volumes to your volume group and create an ext3 partition (of 150 GB, for instance), use **Make share as FTP** and configure *ftpgroup* as the primary group (PG); allocate Read Write (RW) access to *ftpgroup*. Under the host